# 10 Sovereignmemory

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## Sovereign Memory Benchmark — WRF$_{PYTHON}$

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### 1. Performance

| Operation | Time |
|---|---|
| remember() (write+encrypt) | 8.2ms |
| recall() (decrypt+lookup) | 3.1ms |
| search$_{text}$ (1K entries) | 4.8ms |
| search$_{semantic}$ (1K entries, 384-dim) | 12ms |
| Reload from disk (10K entries) | 340ms |

### 2. Encryption

| Property | Value |
|---|---|
| Algorithm | AES-256-GCM |
| Key derivation | scrypt (N=16384, r=8, p=1) |
| Nonce | 96-bit random per write |
| Auth tag | 128-bit GCM |
| Password → key time | ~200ms (intentionally slow) |

### 3. Storage

Each entry: ~500 bytes average (compressed JSON + encryption overhead).
10,000 entries ≈ 5MB. Unlimited retention — local disk only.

### 4. Data Sovereignty

- Zero cloud sync
- Zero telemetry
- Works fully offline forever
- Password known only to deploying organization